

## CIVIL / OPEN RESEARCH PROJECT

# InSAR-Assisted Landslide Early-Warning Prototype

Prototype a geospatial workflow combining satellite deformation signals, rainfall and terrain features to prioritize slopes for civil-engineering inspection.

| TEAM             | DURATION          | FORMAT            |
|------------------|-------------------|-------------------|
| Up to 3 students | 14-week prototype | Mentored research |

## Problem

Slope failures can develop through subtle ground movement before visible collapse. Civil teams need interpretable risk indicators that combine remote sensing with local rainfall and terrain evidence.

## Method

Students will process open Sentinel-1 interferometric products, align them with rainfall and digital elevation data, build an explainable susceptibility model and design a field-verification checklist. The prototype will not issue public safety warnings.

## Research objectives

- Build a reproducible geospatial preprocessing notebook
- Compare rule-based and machine-learning risk scoring
- Quantify uncertainty caused by vegetation and decorrelation
- Produce an inspection-priority map with responsible-use guidance

## Skills and preparation

- GIS or remote sensing
- Python geospatial tools
- Civil/geotechnical fundamentals
- Risk communication

## Expected deliverables

- Documented geospatial pipeline
- Explainable risk-scoring model
- Interactive inspection-priority map
- Validation and limitations report

## Indicative timeline

|             |                                  |
|-------------|----------------------------------|
| Weeks 1-2   | study-area and data audit        |
| Weeks 3-6   | InSAR and terrain processing     |
| Weeks 7-10  | modelling and uncertainty        |
| Weeks 11-14 | validation and engineering brief |

## Responsible-use note

This brief describes a staging research concept. It does not promise funding, credit, authorship, laboratory access, safety clearance or selection. The faculty lead must approve the final protocol, data access, ethics and safety requirements before work begins.

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